Project Phase II Report

Reducing False Negatives in Breast Cancer Diagnosis Using a Hybrid SMOTE + Cost-Sensitive XGBoost Framework

ECOM 6306 – Supervised Machine Learning

Computer Engineering Department – IUG

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1. Problem Definition

In medical diagnosis, particularly in **breast cancer detection**, machine learning classifiers often face **severe data imbalance**, where the number of healthy samples (majority class) greatly exceeds the number of cancer-positive samples (minority class).
This imbalance causes traditional models such as Random Forest or Logistic Regression to become biased toward the majority class, resulting in **high accuracy but low Recall** for the minority (disease) class.

Although **SMOTE (Synthetic Minority Oversampling Technique)** is widely used to generate synthetic samples for the minority class, it has significant **limitations** when applied to **high-dimensional medical datasets** such as histopathology images or gene expression data.
SMOTE can generate **synthetic noise** — unrealistic or overlapping samples — that degrade model performance and sometimes worsen classification results.

On the other hand, **cost-sensitive learning** assigns higher penalties to misclassifying minority-class samples (false negatives), but it struggles to capture complex non-linear boundaries in high-dimensional spaces.

Hence, a **hybrid approach** that combines **data-level resampling (SMOTE)** and **algorithm-level cost weighting (Cost-Sensitive XGBoost)** may overcome the weaknesses of individual methods.
The problem, therefore, is to **design, implement, and evaluate** such a hybrid framework and test whether it **reduces false negatives** and **improves Recall** for breast cancer diagnosis using imbalanced medical data.

2. Research Question

> To what extent can a hybrid approach combining SMOTE and cost-sensitive XGBoost improve minority-class Recall and reduce false negatives in breast cancer diagnosis compared to using SMOTE or cost-sensitive learning individually?

3. Research Objectives

1. **To analyze** why current SMOTE-based and cost-sensitive models fail to achieve high Recall in highly imbalanced breast cancer datasets.
2. **To develop** a hybrid learning framework integrating SMOTE with cost-sensitive XGBoost for improved detection of cancer-positive cases.
3. **To compare** the hybrid approach against traditional methods (SMOTE-only, cost-sensitive-only, and baseline models) using metrics such as Recall, F1-score, and AUC-ROC.
4. **To evaluate** whether the hybrid method effectively reduces false negatives without significantly lowering precision.
5. **To identify** the key factors (imbalance ratio, synthetic noise, cost weighting) influencing model performance.

4. Expected Outcome

1. A validated **hybrid model** that significantly improves Recall for minority (cancer-positive) cases.
2. Demonstrated reduction in false-negative predictions, leading to safer diagnostic outcomes.
3. A set of empirical guidelines for applying hybrid imbalance-handling techniques to other medical domains.
4. A foundation for a future experimental phase (Phase III) where real datasets such as Breast Cancer Wisconsin (Diagnostic) from UCI or MIAS mammography dataset can be tested.

5. Justification

False negatives in cancer diagnosis represent one of the most critical risks in clinical AI systems — a missed diagnosis may delay treatment and threaten patient survival.
Improving Recall for minority disease classes is therefore **not only a technical challenge but an ethical and clinical necessity**.
The hybrid approach (SMOTE + cost-sensitive XGBoost) directly targets this issue by **balancing the data distribution** and **teaching the algorithm to prioritize rare but vital cases**.
This study contributes to building **trustworthy, fair, and reliable AI diagnostic tools** for healthcare.

6. References

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